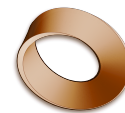


Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics



Planning an Investigation

Homework 1 · Criterion B practice

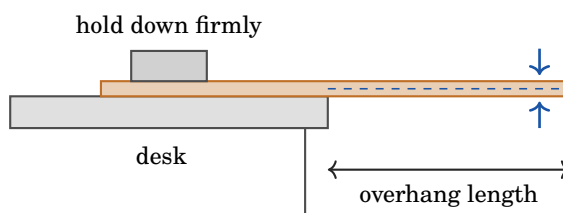
Due: Thursday 24 September

 **About one hour.**

You are about to plan and carry out the **pendulum investigation** in class, and it is a **summative** — so every word of it is written in the lesson, in your own words, from your own data. **None of it goes home.**

So this packet practises the *planning* — **Criterion B** — on somebody else's experiment. Get good at the writing here, and when you sit down in class with your own investigation the writing is the easy part. **Keep your Criteria B and C sheet beside you: each section below is one row of it.**

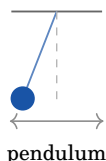
Meet the experiment. Kit and Sam hold a metre rule flat on a desk with part of it over the edge, press the desk end down hard, and twang the overhanging end. It bounces up and down — it **oscillates**. They want to know whether the **length of the overhang** changes **how fast it oscillates**.



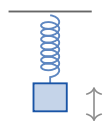
Reading — oscillations and frequency

An **oscillation** is a repeating back-and-forth movement about a middle position. The middle is where the object sits when nothing disturbs it, and an oscillating object keeps passing back through it.

You have met more oscillations today than you realise — every one of them the same idea wearing a different coat.



pendulum



mass on a spring



twanged ruler



guitar string



your vocal cords

5. Kit wrote three possible questions. **Tick** the one that is testable, and say what is wrong with the other two.

✓	Kit's question	What is wrong with it?
	Is a long ruler better than a short one?	
	How does the length of the overhang affect the frequency of the ruler?	
	Why do rulers oscillate?	

6. Rewrite this so it is testable: *“Does the size of a drum change the sound?”*

CRITERION B • strand ii The hypothesis

A **hypothesis** is not a guess — it is a **prediction with a reason attached**, and the reason must be physics.

I think _____ **will** _____
the prediction

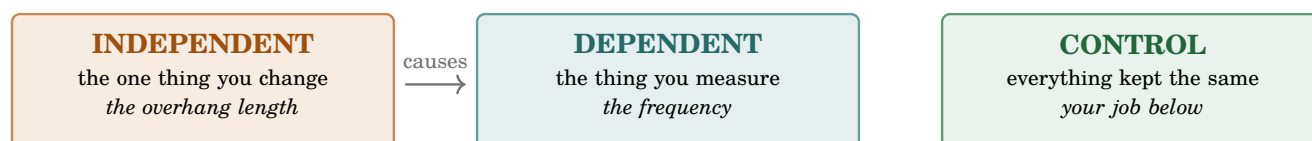
because _____

↑
the marks are here

7. Kit wrote: *“I think a shorter overhang will oscillate faster.”* That is a prediction and nothing more — on the criteria sheet it sits in the bottom band. Add a reason, starting with *because*.

CRITERION B • strand iii Variables, and the data you will collect

One thing changed, one thing measured, everything else held still.



8. Name **three** things Kit must keep the same, and say what would go wrong if each one changed. Then answer the question underneath — the criteria sheet asks for **sufficient** data, not just data.

	Kept the same	What would go wrong if it changed
1		
2		
3		

Kit plans to test **two** overhang lengths, **once each**. Why is that not enough, and what should Kit do instead?

CRITERION B • strand iv The method

Numbered steps a classmate could follow **without asking you a single question**, plus equipment and one safety point.

9. Kit's method has **four** faults. Write what is wrong beside each step.

Kit's method	What is wrong with this step?
1. Put the ruler on the desk.	
2. Twang it.	
3. Time it with a stopwatch.	
4. Write down the answer.	

10. Now write the method properly — five or six numbered steps. A classmate must be able to follow it without asking you anything.

11. A method is not finished until a reader knows what to fetch and what to watch out for. Complete both boxes.

- **Equipment** — everything a classmate would have to collect *before* starting. Be specific: *a metre rule*, not *a ruler*. Remember Kit has to measure the overhang as well as time it. One item per line.
- **Safety point** — name something that could genuinely go wrong in *this* experiment, and say what you would do about it. **“Be careful” is not a safety point** — it names no hazard and changes nothing.

Equipment	Safety point — the hazard, and what you would do

Before you hand this in

Criterion B is written **before you touch the apparatus**. Tick each row.

✓	Strand	Have you...
	B i	...written a question naming what you change <i>and</i> what you measure?
	B ii	...predicted an outcome <i>and</i> given a physics reason after the word <i>because</i> ?
	B iii	...named the independent, the dependent and at least three controls — and said how many readings and how many repeats?
	B iv	...written numbered steps a classmate could follow unaided, with equipment and one safety point?

The five words that decide your level

Your criteria sheet uses the same handful of verbs over and over. The strand barely changes between levels — the *verb* does. Learn these five and you can aim at a band deliberately instead of hoping.

states	you wrote it down
outlines	you gave the short version
describes	you said what it is really like
explains	you said <i>why</i>
discusses	you weighed it up, both sides

Words used in this packet

oscillation	a repeating back-and-forth movement about a middle position
one oscillation	all the way there <i>and</i> all the way back, to where it started
frequency	how many complete oscillations happen in one second. Symbol f , unit hertz (Hz)
hertz	one oscillation per second. kHz = 1 000 Hz, MHz = 1 000 000 Hz
hypothesis	a prediction <i>with a scientific reason attached</i>
independent	the one thing you deliberately change
dependent	the thing you measure
control	anything you keep the same so it cannot affect the result
repeat	doing the same measurement again at the same setting